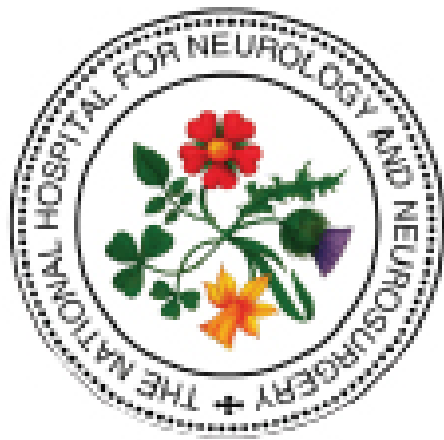




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

Management of Sodium and Water Balance

HYPONATRAEMIA ($\text{Na}^+ < 135\text{mmol/L}$)

Normal plasma sodium =135–145mmo/L
& plasma osmolality 285–295mOsm/kg

Sodium disturbances are common after brain injury and treatment is indicated when there are acute symptoms that pose a risk of significant neurological complications.

The symptoms and signs relate not only to the severity of the sodium disturbance but also to the rate of change of serum sodium concentration.

HYPONATRAEMIA is defined as a serum sodium concentration of $<135\text{mmol/L}$ and it usually develops between 2 and 7 days after injury and is associated with mortality increases of up to 60%. Hyponatraemia creates a gradient across the blood–brain barrier allowing entry of water into the brain and the development of cerebral oedema. The cause of the low serum sodium can be determined by the associated volume status.

Both **CSW** and **SIADH** can occur following brain injury. It is important to distinguish between the two as the treatment differs. For CSW the patient is given fluids and sodium supplementation, while for SIADH the patient is fluid restricted unless the sodium is severe ($<125\text{mmol/L}$) and the patient has significant symptoms.

CSW is mostly seen in SAH and the patient must be volume repleted while treating the hyponatraemia – with isotonic saline for mild to moderate cases, hypertonic saline (1.8%) for moderate to severe cases. In some cases the use of fludrocortisone is also advocated.

Following confirmation of **SIADH**, the aim is to restrict the intake of water. Important drug-related causes include diuretics (particularly thiazides) and anticonvulsant drugs (particularly carbamazepine).

The rate of serum sodium improvement in both CSW and SIADH should be gradual (8–10mmol/24h) in most circumstances to avoid complications associated with rapid osmolar shifts (cerebral oedema and osmotic demyelination syndrome). Severe hyponatraemia (irrespective of the cause) may lead to neurological complications and requires prompt management with intense monitoring.

CONSIDER TREATING SERUM SODIUM $\leq 130\text{mmol/L}$

